

Krish Patel

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Toronto, ON (Open to Relocation / Remote)

EDUCATION

Toronto Metropolitan University

Master of Engineering (M.Eng.), Electrical and Computer Engineering
Concentration: Artificial Intelligence

Toronto, ON
Sept 2026 – Present

University of Western Ontario

Bachelor of Science, Specialization in Computer Science

Relevant Coursework: Operating Systems, Distributed Systems, Data Structures & Algorithms, Database Systems

London, ON
July 2026

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, LangGraph, Spring Boot, Node.js, React, Next.js, REST APIs, TensorFlow, PyTorch, Scikit-learn

Testing & Automation: Pytest, Unittest, Robot Framework, Regression Testing, Test Automation, Integration Testing, TDD

Cloud & DevOps: Git, GitHub Actions, CI/CD Pipelines, Docker, Kubernetes, Jenkins, AWS (EC2, S3), Linux (CLI)

Concepts: OOP, Software Quality Assurance, Debugging, Agile/Scrum, IBM Z (Certified), Machine Learning, NLP

WORK EXPERIENCE

Software Engineering Fellow

Headstarter AI

May 2025 - Aug 2025
New York (Remote)

- Developed **Python**-based test automation scripts to streamline regression testing, improving test cycle efficiency by **30%**.
- Reduced **CI/CD** deployment errors by **15%** by refactoring automated testing scripts and pipeline configurations alongside senior engineers from **Amazon** and **Bloomberg**.
- Built and shipped **3+ full-stack AI applications** serving **500+ users** with **React** and **Node.js**, achieving **98% unit test coverage** with Jest across all core service modules.
- Authored internal API documentation and sprint artifacts, cutting engineer onboarding time by **20%** and improving delivery velocity across a distributed cross-functional team.

Data Engineer Intern

ZippiAI Inc.

Jun 2024 - Aug 2024
San Francisco (Remote)

- Architected scalable **Python**-based ETL automation pipelines using **SQL**, increasing user engagement by **35%** through data-driven feature prioritization.
- Built **Python** automation scripts with integrated unit testing to streamline client onboarding and CRM synchronization, saving **10+ hours** of manual work weekly while maintaining **100% data integrity**.
- Optimized **PostgreSQL** schema designs and complex SQL queries for a **3,000+ client** database, reducing API response latency by **15%** through indexing and OOP-based query abstraction.

PROJECTS

AI Log Detective & SRE Observability Platform (Sentinel)

Python, LangGraph, FastAPI, PostgreSQL

- Engineered a multi-agent SRE observability platform with stateful **LangGraph** workflows to automate root cause analysis, reducing system downtime by **40%**.
- Built a high-concurrency ingestion engine using **Python Asyncio** to process **100+ simultaneous** telemetry feeds and generate RCA reports in under **5 seconds** end-to-end.
- Integrated **LLM agents** via **MCP SDK** to map anomalous log patterns to source code locations, automating patch generation and cutting manual debugging time by **30%**.

Real-Time ML Monitoring Engine (Aeon)

Python, Scikit-learn, TensorFlow, PostgreSQL

- Built a supervised ML pipeline to detect equipment corrosion from sensor telemetry using **Scikit-learn** and **TensorFlow**, improving predictive accuracy by **20%** over baseline.
- Processed **1TB+** of structured sensor data with optimized **PostgreSQL** schemas and indexed SQL queries, reducing training data retrieval time by **60%** and enabling real-time inference at scale.
- Implemented model drift detection scripts triggering automated retraining via cron orchestration, sustaining a **15% improvement** in equipment reliability over a **6-month** production window.

Automated Test Framework & Validation Utility

Python, Pytest, Git, Test Automation

- Designed and built an open-source test automation framework in **Python** utilizing object-oriented principles to execute end-to-end testing scripts across multiple software modules.
- Implemented automated log parsing, data sanitization scripts, and edge-case verification modules to intercept software execution anomalies with **100% precision**.
- Configured a version-controlled test pipeline using **Git** to log test iteration results, run programmatic debugging steps, and generate automated test execution reports.